Lab 01: Basic LAN Setup

2911 Router · 2x 2960 Switches · PC0 & PC1

Lab #01	**Topic:** Networking Fundamentals	**Tool:** Cisco Packet Tracer	**Cert:** CompTIA Network+

1. Lab Objectives

- Build a Local Area Network using two Cisco 2960 switches, a 2911 router and two PCs.
- Configure static IP addresses on both PCs with correct default gateways.
- Bring up router Gigabit interfaces and verify Layer 3 end-to-end connectivity.
- Use **ping** to confirm that PC0 and PC1 can communicate through the router.
- Relate to CompTIA Network+ objective 2.2: Compare and contrast networking devices.

2. Network Topology

The topology shows a Cisco 2911 router (Router0) at the top connected via Gig0/0 to Switch0 (2960-24TT) and via Gig0/1 to Switch1 (2960-24TT). PC0 connects to Switch0 on Fa0/1, and PC1 connects to Switch1 on Fa0/2.

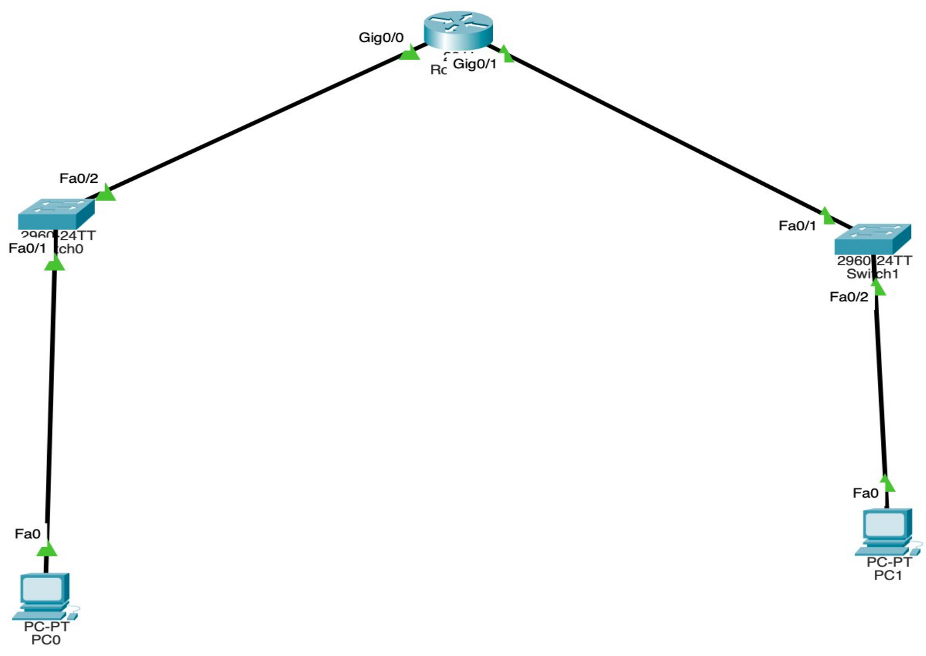

Figure 1 – Basic LAN topology: Router0, Switch0, Switch1, PC0, PC1

Device	Model	Interface	IP Address	Subnet Mask	Default GW
Router0	2911	Gig0/0	192.168.1.1	255.255.255.0	—
Router0	2911	Gig0/1	192.168.2.1	255.255.255.0	—

Switch0	2960-24TT	—	—	—	—
Switch1	2960-24TT	—	—	—	—
PC0	PC-PT	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	PC-PT	Fa0	192.168.2.10	255.255.255.0	192.168.2.1

3. Step-by-Step Configuration

3.1 Router0 – Gigabit Interfaces

```
Router> enable
Router# configure terminal
Router(config)# hostname Router0
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# ip address 192.168.1.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface GigabitEthernet0/1
Router0(config-if)# ip address 192.168.2.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# exit
Router0# write memory
```

3.2 PC IP Configuration

PC0 – IP	192.168.1.10
PC0 – Mask	255.255.255.0
PC0 – Gateway	192.168.1.1
PC1 – IP	192.168.2.10
PC1 – Mask	255.255.255.0
PC1 – Gateway	192.168.2.1

4. Verification

```
PC0> ping 192.168.2.10        ! PC0 to PC1 through router
PC0> ping 192.168.1.1         ! PC0 to its gateway
Router0# show ip interface brief  ! Verify Gig0/0 and Gig0/1 are up/up
Router0# show ip route            ! Check connected routes
```

5. Key Takeaways

- The Cisco 2911 router uses GigabitEthernet interfaces (not FastEthernet like the 1841).
- Each router interface connects a separate subnet — two interfaces = two subnets here.
- Switches (2960) operate at Layer 2 and forward frames by MAC address, requiring no IP.
- Without a default gateway configured on the PCs, cross-subnet pings will fail silently.
- Always verify interfaces are 'up/up' with **show ip interface brief** before troubleshooting.